

76 SOPs, 5 CHAPTERS, 14 ROLES

— ONE PLATFORM, ONE SIGN-OFF CHAIN, ONE SCHEDULE.

Source: the freshly scanned **ILCO SOPs digital scan** (04 May 2026) covering documentation handling, legal compliance, cannabis cultivation, general facility & product handling, and security. This plan maps every SOP to the role that owns it, flags the digital gap, and sequences a 4-phase build that turns paper procedures into tap-and-sign workflows in TnT-ZA.

⚠ **SOURCE-OF-TRUTH CONFLICT TO RESOLVE BEFORE ANY ARTEFACT SHIPS**

The 2020 SOP cover sheets list **Authoriser: Coenraad Venter · Approver: Ilse Venter**. Current canonical roster (`seed-real-staff.ts`): **Ilse = AR · Coenie = DAR · Berne Swart = RP**. Either the 2020 SOPs use generic "authoriser/approver" labels (not regulatory AR/RP titles), or the role assignment has flipped since 2020 and the SOP footers are stale. **Confirm**; we re-issue or re-map accordingly.

SOPS IN SCOPE

76

across 5 chapters

ALREADY DIGITAL

~21

all of Chapter 3 Cultivation

HALF-BUILT

3

primitives need generalising

STILL PAPER

~52

Ch 2, Ch 4 events, Ch 5/6 logs

01 FIVE PRIMITIVES EVERY SOP REUSES

Build these once in Phase 0 → every form below inherits them. Without these, every SOP re-implements the wheel.

PRIMITIVE	WHAT IT DOES	REUSED BY
Sign-off chain widget	3-row chain: Compiler → Authoriser → Approver. Locks on signature, hash-chained to <code>AuditLog</code> . Already exists for SMF sections — generalise to any SOP record.	76 / 76
Form 6.1 · Amendment History	Append-only revision table on every SOP: revision no, what changed, who, when, why.	76 / 76
Form 6.2 · Distribution List	Numbered authorised-copy roster with "AUTHORISED COPY" green stamp + retrieve-on-supersession hook.	76 / 76
Logbook primitive	Append-only entry stream with optional dual-sign + photo + per-entry hash. One component, many tables.	18 SOPs
Schedule engine	Cron + ticket spawner with SLA. Cadence becomes a row in <code>ScheduleRule</code> ; firing creates a ticket; Maestro surfaces it as "due / overdue / coming up".	15 SOPs

02 PER-ROLE DIGITAL UX · WHAT EACH PERSON TAPS

One row per role. Aligned to the canonical roster from `seed-real-staff.ts` + the agent's per-SOP responsibility map.

ROLE · PERSON	OWNS / SIGNS	TAP ACTIONS IN CHAT (NO TYPING)
AR · Ilse Venter	All 2-LEG-* approvals · 4-FPH-007 / 012 / 028 (witness) · 1-DOC-001 / 002 / 003 final approve	What's waiting on AR? · Approve & freeze · SAHPRA licence T-90 · Annual director declaration
DAR · Coenie Venter	2-LEG-009 deputy duties · co-signs 2-LEG-003 / 006 / 011 in AR absence	DAR queue · Drift triage · Ghost-act as RP · Escalations to AR
RP · Berne Swart	2-LEG-010 / 015 · 4-FPH-015 FP release · 4-FPH-025 S6 register · 4-FPH-027 month-end · 4-FPH-028 / 029	Stale SMF sections · Draft with AI · FP release queue · S6 in/out today · ADR · raise · Month-end checklist
HoC · Lourens Eksteen	All 3-CUL-001..010 · 3-INV-008 / 011 · IPM scouting	Today's plant tasks · Harvest window 14d · IPM weekly scout · Log feed / pH
FM · Renae Purdon	4-FPH-008 / 009 / 010 / 021 / 026 · 4-SEC-009 / 010 · co-signs 2-LEG-001	Greenhouse status · Overdue maintenance · Probe health · Genset weekly test · Monthly expiry sweep
ProcMgr · Jeanette Ferreira	4-FPH-013 / 015 / 017 / 018 / 019 / 020 · signs trim / dry / cure / pack stage records	Batches by stage · Today's trim queue · Dry-room weights · Cure jar rotation · Stability pulls due
QA Inspector · Keke	4-FPH-002 / 003 / 005 / 014 / 015 / 017 / 020 / 022 / 023 · signs all 3-CUL stage records	Open deviations · COAs awaiting review · Open CAPAs by age · Trend last 30d · Raise deviation
Security · Sipho [⚠]	6-SEC-001 / 002 / 003 / 005 · 4-SEC-006 / 007 / 008	Overnight access events · Perimeter status · Log perimeter walk · Visitor sign-in · CCTV / alarm weekly test
IT · Calvin Green	6-SEC-004 only (information-system security)	IT ticket queue · Quarterly access review due · Printer queue · Network health
Admin / HR · Loraine	2-LEG-011 / 012 / 013 / 014 · 1-DOC-004 organogram	Training expiries 30d · Today's absences · Onboarding pipeline · Archive-purge candidates (5y)
Trimmer (1 of 25)	3-CUL-010 execution + per-batch trim log	Today's job · Log weight · Raise issue · End shift
Cultivator / Nursery / Irrigation	3-CUL-001..009 execution + 3-INV-008 / 011 logs	Today's plant tasks · Log nutrient mix · Logtag download · Raise pest sighting
Compiler (unmapped)	Compiler signature on every SOP (76 / 76) · 1-DOC-001 / 002 / 003 ownership	SOPs due for 3-yr review · New SOP request queue · Distribution list audit
SAHPRA Inspector	Reads everything · writes nothing	Show current SMF · Replay how a section was written · Verify audit hash chain · Export section + sign-offs

[⚠] Sipho appears in the previous staff memory but not in the current `seed-real-staff.ts` . Verify before going live.

03 PER-SOP DIGITAL READINESS · THE GAP MAP

Five non-exclusive flags per SOP. Most regulated forms carry 2-3 flags simultaneously.

PAPER still pen-on-form · SCHEDULED daily / weekly / monthly cadence · EVENT triggered by deviation / scan / sample · SIGNED regulated signature chain · LOGBOOK append-only continuous log

CHAPTER	SOP COUNT	DOMINANT PATTERN	FLAG MIX
1 · Documentation	4	SOP lifecycle (create / distribute / version / organogram)	PAPER SIGNED EVENT
2 · Legal & QMS	17	Heavyweight regulator-facing forms (TQCRF, Recall, SIA, ADR, SMF, training matrix)	SIGNED SCHEDULED EVENT
3 · Cultivation	17	Per-stage plant lifecycle + Logtag T & H + nutrients · already digital	EVENT LOGBOOK SIGNED
4 · Facility & Product Handling	28	Deviation / CAPA spine + monthly cadences (expiry, month-end, pest) + S6 register	EVENT SCHEDULED SIGNED LOGBOOK
5 / 6 · Security	10	Daily / weekly logbooks (perimeter, vault, visitor, genset) + IT access review	LOGBOOK SCHEDULED SIGNED

Each phase ships independently. Phase 0 is mandatory; Phases 1–4 can be re-ordered if business priority changes.

PHASE 0 · FOUNDATION

The 5 platform primitives. Everything downstream inherits these. Without Phase 0, every form re-builds the wheel.

1 SPRINT

DELIVERABLE	DETAIL
Prisma models	SopRecord · SopVersion · SopDistribution · SopSignature · LogbookEntry · ScheduleRule
React widget	<SignOffChain> · 3-row Compiler / Authoriser / Approver with signature lock
React widget	<AmendmentHistory> + <DistributionList> auto-attached to every SOP entity
React widget	<Logbook> with append-only enforcement, dual-sign + photo support
Schedule engine	Cron daemon writes tickets · Maestro picks them up · "due / overdue" surfaces in chat

PHASE 1 · HIGH-VOLUME LOGBOOK WINS

Biggest daily-friction wins. Replaces pen-and-clipboard on the highest-touch logs.

1 SPRINT

SOP	SURFACE	OWNER
4-FPH-025 · S6 register	Tap-to-log in / out · dual sign · QR on jar	RP
6-SEC-005 · Vault entry	Tap-to-log entry / exit · dual sign · CCTV link	Security
4-SEC-007 · Visitor book	iPad sign-in at gate · photo · auto-leave on exit	Security
4-FPH-019 · Cold chain	Logtag CSV auto-ingest · breach → ticket	ProcMgr
3-INV-008 · Logtag T & H	Same Logtag pipeline · breach → SMF drift	HoC + RP

PHASE 2 · EVENT-DRIVEN CORE

The regulated event spine — every other form chains off these.

2 SPRINTS

SOP	SURFACE	OWNER
4-FPH-002 · Deviation	"Raise deviation" button on every screen · auto-classify · ticket	All
4-FPH-003 · CAPA	Linked from any deviation · root cause + closure SLA	QA
4-FPH-005 · T & H breach	Auto-spawned from Logtag breach (Phase 1)	RP
4-FPH-007 · Change control	Form + AR final approval · feeds 1-DOC-003	AR
2-LEG-003 · TQCRF	Dedicated complaints form · quarantine link · quarterly review widget	AR
2-LEG-004 · Recall	18-field post-recall form · class I / II / III workflow · SAHPRA notify	AR + RP
2-LEG-015 · ADR	Pharmacovigilance form · SAHPRA timeline tracker	RP



PHASE 3 · SCHEDULED CADENCE

1 SPRINT

Calendar engine pays off. Daily / weekly / monthly / annual cadences become tickets, not memory tax.

CADENCE	SOPS	SURFACE
Daily	6-SEC-001 perimeter · 4-FPH-009 hygiene log	Mobile checklist
Weekly	6-SEC-002 CCTV / alarm · 4-SEC-010 genset · 4-FPH-008 housekeeping	Friday widget
Monthly	4-FPH-010 pest · 4-FPH-021 expiry sweep · 4-FPH-026 spillage kit · 4-FPH-027 month-end · 4-FPH-018 stability pull	Monthly cockpit
Quarterly	2-LEG-003 complaints review · 6-SEC-004 IT access review	RP / IT widget
Annual	2-LEG-006 SIA · 2-LEG-005 vendor · 2-LEG-007 courier · 4-FPH-004 risk · 4-SEC-009 fire · 2-LEG-001 director declaration	Year-plan widget
Periodic	1-DOC-001 3-yr SOP review · 1-DOC-003 5-yr archive purge · 2-LEG-001 90-day licence renewal	Compiler cockpit



PHASE 4 · HEAVYWEIGHT REGULATOR FORMS

2 SPRINTS

The 5 SOPs that carry 80% of regulated event flow. Polish + dedicated wizards.

SOP	WHY HEAVYWEIGHT
2-LEG-006 · SIA	9-section, 59-page audit checklist · digital wizard with section-by-section sign-off · 5-yr retention
2-LEG-004 · Recall	18-field post-recall form + class I / II / III decision tree + SAHPRA notify chain
2-LEG-003 · TQCRF	Full complaints form with quarantine + RCA + CAPA threading + quarterly review
4-FPH-002 / 003 · Deviation + CAPA	From Phase 2 · gets full polish: trend analytics, SLA escalation, cross-batch linking
4-FPH-015 · FP release	Per-batch RP release with COA gate · S6 register integration · destruction handoff

Honest baseline before we cut Phase 0 tickets.

ALREADY BUILT

Cultivation surfaces (Ch 3)	21 SOPs digitised across the cultivation dashboard
RP / DAR / AR / QA dashboards	Sign-off queue · SMF drift · stale-section radar
SMF Composer (Opus 4.7)	Drafts regulator-grade text with hash-chained audit
SMF Drift Daemon	Catches operational changes that invalidate signed sections
Tickets system	Substrate for everything · the operating layer
AuditLog + AICall	Hash-chained log of every action and every Claude call

HALF-BUILT · NEEDS GENERALISING IN PHASE 0

Sign-off chain	Works for SMF sections; not yet generalised to all SOPs
Logbook primitive	Cultivation logs done; not generalised to S6 / vault / visitor / cold chain
Schedule engine	Drift daemon exists; cadence-based scheduler not yet

NOT BUILT · THE MEAT OF PHASES 1–4

All 18 LOGBOOK SOPs	Need the generalised Logbook primitive (Phase 0) + per-SOP wiring (Phase 1)
All 25 EVENT-DRIVEN SOPs	Need the deviation / CAPA event spine (Phase 2)
All 15 SCHEDULED SOPs	Need the cron + ticket engine (Phase 3)
Chapter 2 Legal forms	TQCRF, Recall, SIA, ADR, training matrix wizards (Phase 4)
Compiler's SOP-version cockpit	3-yr review queue, distribution list audit, change control feed

OWNER SIGN-OFF NEEDED ON:

1. **AR / RP / DAR mapping conflict.** 2020 SOPs say Coenie = Authoriser, Ilse = Approver. Current canonical: Ilse = AR, Coenie = DAR, Berne = RP. Confirm so we re-issue or re-map.
2. **Compiler line on the SOP cover sheet** — provision a dedicated user, or assign an existing role-holder to wear that hat?
3. **Sipho on Security** — confirm still on staff, then add to `seed-real-staff.ts` .
4. **Phase ordering** — Phase 1 (logbook wins) before Phase 2 (event spine) is the proposal. Or do regulator-grade event forms (TQCRF / Recall / SIA) come first?
5. **Verbatim SOP body reads** — for SOPs beyond the 6 read in this audit, flags are inferred from titles + the universal template. Want a verbatim read of the remaining 54 PDFs before tickets get cut?